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A Cloud-Powered Meta-Heuristic Model for Enhanced Efficiency in Intermittent Gas Lift Operations: A Case Study

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Abstract

Intermittent gas lift (IGL) wells connected to a centralized high-pressure gas source often encounter injection interference, leading to header pressure fluctuations and liquid surges. This study introduces an optimization model using three meta-heuristic algorithms to stagger injection cycles and minimize peak instantaneous gas demand. Built on an Amazon Web Services (AWS)-based architecture, this solution enables production engineers to optimally manage IGL well cycles, reducing peak instant gas load and minimizing production losses.

The combined use of three meta-heuristic algorithms, Simulated Annealing, Particle Swarm Optimization, and Genetic Algorithm, provides an effective approach to optimizing time cycle staggering in intermittent gas lift wells. Simulated Annealing's convergence capabilities, Particle Swarm's exploration strength, and Genetic Algorithm's adaptability make this combination well-suited to handle the NP-hard nature of time cycle staggering. To reach a near-optimal outcome, the exploration space of each algorithm is expanded, significantly increasing computational demands. An AWS-based architecture is implemented to address the computation challenge and ensure a production ready application for scalable performance.

A Manual staggering of time cycles becomes increasingly impractical as the number of wells in a field grows. This application automates the process, achieving near-optimal time cycle staggering solutions despite the NP-hard complexity of the problem. Leveraging AWS-powered architecture, this application enables production engineers to enhance intermittent gas lift efficiency by staggering gas injection cycles, thereby stabilizing the pressures in the lift gas network and reducing production losses. In ONGC's onshore fields, this solution demonstrated a substantial reduction in peak instantaneous gas fluctuations (~40-50%), lowering production losses by minimizing instantaneous peak gas fluctuations.

This work introduces a novel application of meta-heuristic algorithms-based optimization and cloud computing to enhance intermittent gas lift operations. By automating the staggering gas injection cycles in IGL wells, the application significantly reduces peak instantaneous gas fluctuations and pressure fluctuations (up to 50%). This scalable AWS-powered solution offers production engineers a powerful tool to boost operational efficiency and minimize production losses, setting a new benchmark for artificial intelligence-driven approaches in gas lift management.

Introduction

Intermittent Gas Lift (IGL) is a widely adopted artificial lift technique used in mature oil fields where declining reservoir pressures necessitate gas-assisted production. In IGL systems, wells operate in cyclic modes alternating between gas injection and shut-in periods to lift slugs of accumulated liquid. The intermittent nature of this process results in high instantaneous gas demand during short injection windows. When multiple IGL wells are connected to a common high-pressure gas source, concurrent gas injections can create severe instantaneous gas surges, leading to pressure drops in the lift gas header network and resulting in inefficient slug lifting, missed cycles, and associated production losses.

This challenge becomes especially pronounced in fields with a large number of IGL wells. Even when each well's daily gas consumption remains within the field's gas availability limits, the transient peaks during overlapping injection cycles often cause operational inefficiencies. Traditional methods, such as manual staggering of injection start times, become infeasible as the well count increases and their individual cycle characteristics differ. Moreover, these overlapping peaks not only affect production from the individual wells but also destabilize the broader network and reduce operational efficiency.

To address these challenges, this paper introduces a robust and scalable cloud-powered optimization system for minimizing peak instantaneous gas demand in IGL networks. The system leverages the sequential application of three well-established meta-heuristic algorithms, Simulated Annealing (SA), Particle Swarm Optimization (PSO), and Genetic Algorithm (GA), to stagger the injection cycles across wells. Each algorithm contributes unique exploration or convergence strengths, enabling the combined framework to efficiently explore the solution space and converge toward a near-optimal injection schedule.

The proposed solution is deployed on a scalable, serverless architecture using Amazon Web Services (AWS), enabling rapid computation for small-scale networks and cost-effective asynchronous processing for large-scale fields. Field-scale evaluation has been conducted across three production fields in ONGC, India, Field X (71 wells), Field Y (80 wells), and Field Z (29 wells). In each case, the optimization solution delivered significant reductions in peak instantaneous gas demand by 46%, 41%, and 50% respectively, ensuring smoother header pressure profiles and enhanced production reliability.

Challenges in Intermittent Gas Lift Operations

IGL wells typically operate with a defined cyclic schedule of gas injection and shut-in periods. During the shut-in phase, reservoir fluids accumulate above the operating gas lift valve, forming a liquid slug. When the injection valve opens, high-pressure gas is injected to displace and lift this slug to the surface. The duration of gas injection (ON time) is typically short—ranging from 3 to 6 minutes—while the accumulation (OFF time) spans several multiples of that. Consequently, the instantaneous gas rate required during injection is orders of magnitude higher than the average daily gas allocation.

For example, a well configured with a 60-minute cycle (55 minutes OFF and 5 minutes ON), requiring 5000 m³/day of injection gas, will draw gas at an effective instantaneous rate of 60,000 m³/day during its 5-minute ON period. When multiple wells initiate injection within overlapping windows, the header network experiences extreme transients in demand, far exceeding compressor design limits and destabilizing network pressure.

Even though the overall daily gas allocation is within capacity, these short-term spikes can lower the network pressure below the valve operating threshold, preventing slug lifting and causing full-cycle production losses. Furthermore, overlapping surges from multiple wells can lead to hydraulic instability and slug flow-induced vibrations in the production header, amplifying mechanical wear and reducing system lifespan.

To ensure efficient operations, injection cycles must be carefully staggered such that gas demand is distributed evenly over time. However, with dozens of wells and differing ON/OFF cycles, manual scheduling becomes intractable, and heuristic or optimization-based approaches are required.

Problem Statement and Objective Function

The problem of staggering gas injection cycles in IGL wells can be framed as a combinatorial optimization problem where the objective is to minimize the maximum instantaneous gas rate across the injection scheduling horizon. Each well has a known time cycle (e.g., 55/5, 84/6, etc.), and the starting time for each cycle must be chosen such that the overlap of gas injection windows across wells is minimized.

This optimization falls under the category of NP-hard problems, given the large search space and the discrete, cyclic nature of the decision variables. The goal of the proposed optimization model is not merely to reduce overlaps, but to directly minimize the instantaneous gas demand profile across the entire time horizon, i.e., reduce peak demand and flatten the demand curve.

The objective function used for all three meta-heuristic algorithms is:

Minimize $Max_Instantaneous_Gas_Load(t)$ over the Least Common Multiple (LCM) of all cycle durations
Subject to:

- Fixed cycle duration for each well (ON + OFF times)
- Start time must be within the LCM cycle window
- No modification of cycle length (only start time adjustment allowed)

Each candidate solution is evaluated by simulating the injection profile of all wells over the scheduling horizon and computing the peak instantaneous gas rate. The fitness of each solution is inversely proportional to this peak.

Optimization Methodology Using Sequential Meta-heuristics

To efficiently reach near-optimal solutions in this complex search space, we adopt a sequential hybridization approach that chains three meta-heuristic algorithms: Simulated Annealing (SA), Particle Swarm Optimization (PSO), and Genetic Algorithm (GA). This methodology leverages the unique strengths of each algorithm at different stages of the optimization process:

Simulated Annealing (SA): Initialization & Convergence Zone Mapping

Simulated Annealing is used as the first stage in the optimization pipeline to rapidly escape poor local optima and map out promising regions in the solution space. Its probabilistic acceptance of worse solutions in early iterations allows it to explore broadly and avoid premature convergence.

- Role: Global exploration with controlled randomness
- Benefit: Finds multiple low-cost regions rather than prematurely converging
- Output: Refined set of candidate start times with good spacing

SA outputs a refined population of candidate solutions that serve as a robust starting point for the next phase.

Particle Swarm Optimization (PSO): Intermediate Exploitation

Particle Swarm Optimization is used next to refine the population obtained from SA. Each particle represents a solution vector, and its movement is guided by both personal and global best solutions observed so far.

- Role: Focused exploitation of high-potential zones discovered by SA
- Benefit: Efficient convergence to a near-optimal neighborhood by leveraging swarm intelligence
- Output: Improved solution vectors with reduced peak load

PSO enhances the initial SA-generated solutions by collectively guiding particles toward optimal configurations with minimal overlap.

Genetic Algorithm (GA): Final Tuning and Population Diversity

GA is used in the final stage to inject diversity back into the solution pool and avoid local convergence traps. Chromosome crossover and mutation introduce new candidates, allowing exploration around the PSO-refined solutions.

- Role: Local refinement and diversity recovery
- Benefit: Escapes minor local optima and polishes the final solution
- Output: Final injection scheduling minimizing instantaneous gas load

This structured sequencing ensures that the model first explores widely, then exploits efficiently, and finally fine-tunes the solution. Compared to using any single algorithm alone, this approach consistently delivers better solutions in fewer iterations with lower variance in output quality.

Scalable Cloud-based System Architecture

To operationalize the optimization workflow for production environments, the full system has been implemented on AWS, leveraging its serverless and autoscaling features to deliver reliable performance at low cost. The architecture is designed to be **user-responsive, compute-efficient, and scalable**.

Workflow Design

1. Request Submission:

- Users interact via a web that submits optimization requests through Amazon API Gateway.
- Input includes well data (cycle durations, gas targets, etc.).

2. Logic Routing (Lambda Function):

- A lightweight AWS Lambda function determines whether the request is "small" or "large".
 - If well count < 10: Optimization is executed via the same Lambda function.
 - If well count ≥ 10 : Request is pushed to an Amazon SQS queue.

3. Async Processing (AWS Fargate):

- For large-scale optimization jobs (10 minutes to 2 hours runtime), a containerized application hosted on AWS Fargate processes the request.
- The container runs the three-stage meta-heuristic model sequentially and stores the result in JSON format.

4. Post-processing & Reporting (Lambda + S3 + SES):

- Once optimization is complete, a separate Lambda function is triggered to:
 - Parse and visualize results
 - Generate a report (graphs, stats, summary)
 - Store it in Amazon S3
 - Email it to the user via Amazon SES

5. User Experience:

- Users receive immediate acknowledgment upon request submission.
- Final results are emailed asynchronously within minutes to hours, avoiding UI wait-time.

Cost and Scalability Advantages

- **Elasticity:** AWS Fargate dynamically allocates resources based on job size, eliminating the need to pre-provision infrastructure.
- **Cost Efficiency:** No idle compute costs, pay only for what is used.
- **Serverless Simplicity:** AWS Lambda handles logic routing and reporting without requiring backend servers.
- **Decoupled System:** Using SQS ensures reliability and queuing for burst loads.
- **Global Reach:** Secure, scalable access from any remote location across ONGC assets.

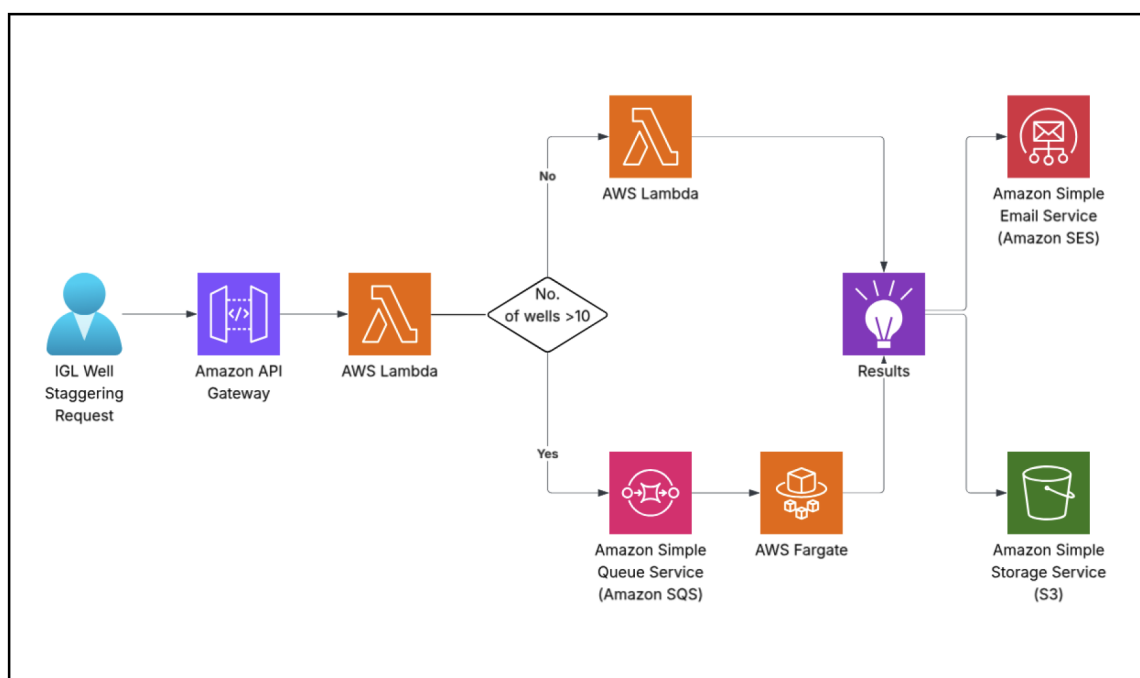


Figure 1—High level system architecture of the solution

This architecture has allowed engineers to apply optimization without requiring specialized hardware, enabling wide deployment across ONGC's production clusters.

Case Studies

To validate the performance and scalability of the proposed cloud-powered optimization framework, comprehensive field-scale studies were conducted across three production clusters managed by ONGC. These sites, referred to as **Field X**, **Field Y**, and **Field Z** for confidentiality, span a diverse operational spectrum in terms of well count, cycle complexity, and network architecture. The objective in each case was to assess the system's ability to reduce peak instantaneous gas demand by generating optimized time cycle staggering schedules for intermittent gas lift wells.

Each field was initially operating under conventional practices involving manual scheduling (excel based), with little optimization of injection overlap. Time cycles for these fields were analyzed and

optimized using the proposed solution, which applies a sequential combination of meta-heuristic algorithms, demonstrating a significant potential reduction in peak gas load across all sites.

Field X – Large-Scale Optimization: 71 IGL Wells

Field X consisting of 71 intermittent gas lift wells connected to a common high-pressure gas manifold. Prior to optimization, operational records indicated 13 simultaneous gas injections occurring at peak congestion intervals, resulting in a maximum instantaneous gas demand of 7.01 LSCMD (Lakh Standard Cubic Meters per Day). These transients consistently stressed the compressor station and contributed to pressure dips that disrupted valve performance.

Using the proposed optimization workflow, injection cycles were staggered to distribute the gas load more evenly across the scheduling horizon. The optimized schedule achieved a reduction in peak instantaneous gas demand to 3.75 LSCMD, a 46.5% improvement over the baseline scenario. The number of concurrent injection events was also lowered from 13 to 10 wells, reducing the likelihood of gas header pressure collapse during overlapping injection phases.

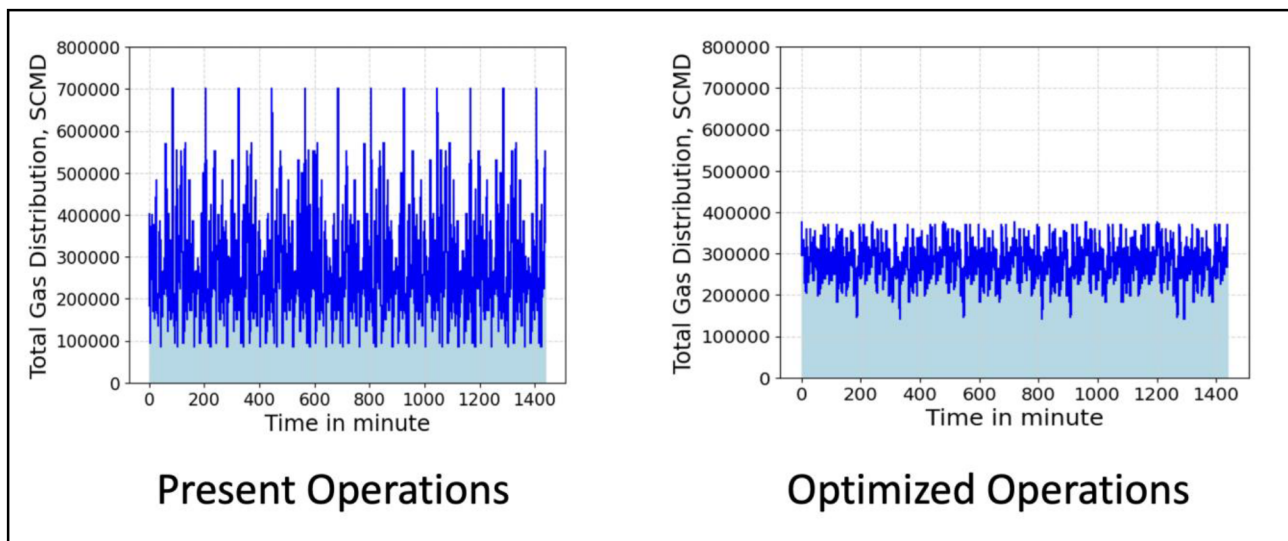


Figure 2—Peak instantaneous gas rate fluctuation with time, for X-field of ONGC, resulting in ~46% reduction in peak instant gas load

Field Y: 80 IGL Wells

Field Y, comprising 80 IGL wells, presented the most computationally intensive test case due to its high well count and overlapping cycle complexity. Under existing operations, the system experienced peak simultaneous injection from 13 wells, with a maximum gas demand of 8.23 LSCMD, resulting in frequent compressor throttling and inefficient gas utilization.

Following application of the optimization system, the maximum simultaneous injection events were reduced to 9 wells, and the peak instantaneous gas demand dropped significantly to 4.81 LSCMD, representing a 41.5% reduction. This decline was especially impactful given the field's tight infrastructure envelope and limited gas throughput capacity.

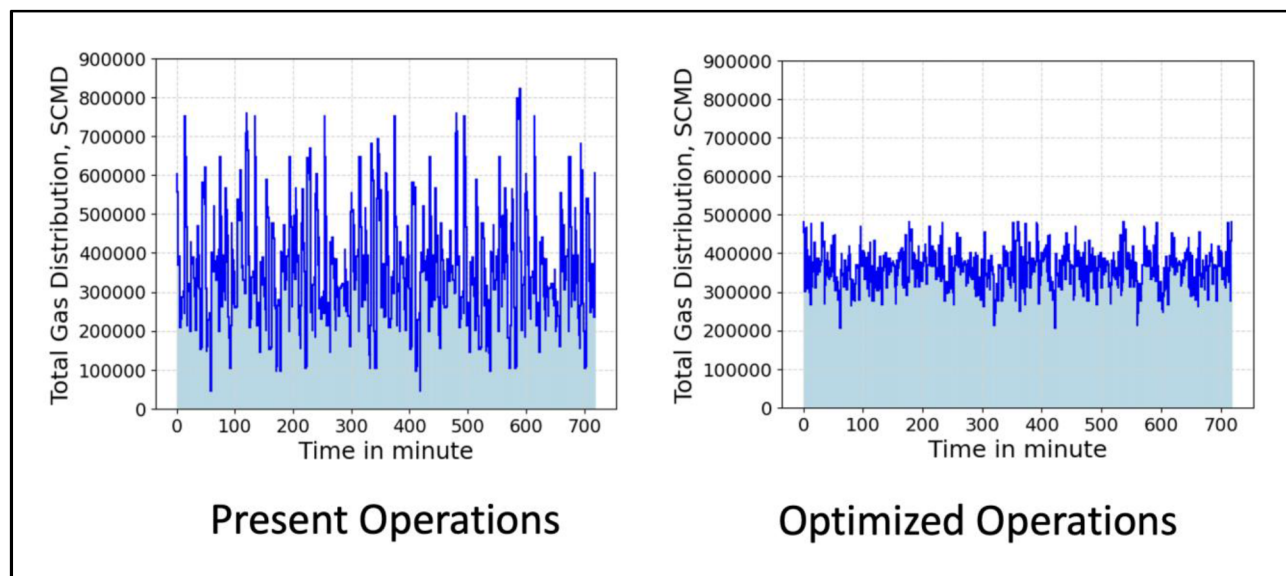


Figure 3—Peak instantaneous gas rate fluctuation with time, for Y-field of ONGC, resulting in ~41% reduction in peak instant gas load

Field Z: 29 IGL Wells

Field Z consisting of 29 intermittent gas lift wells. Though smaller in scope, the field encountered localized header pressure fluctuations due to short bursts of concurrent gas injection, especially when cycle alignment occurred unintentionally across clusters.

Initial assessment showed a peak demand of 4.36 LSCMD under the existing operation, with 4 wells injecting gas concurrently at maximum overlap. While the system was not consistently overloaded, the pressure dips were enough to occasionally prevent lift valve opening, resulting in missed production cycles.

After optimization, the solution produced a time cycle staggering scheme that reduced peak instantaneous gas demand to 2.17 LSCMD, corresponding to a 50.2% decrease. The maximum number of concurrent injections was cut in half to 2 wells, dramatically improving the stability of header pressure profiles across the manifold.

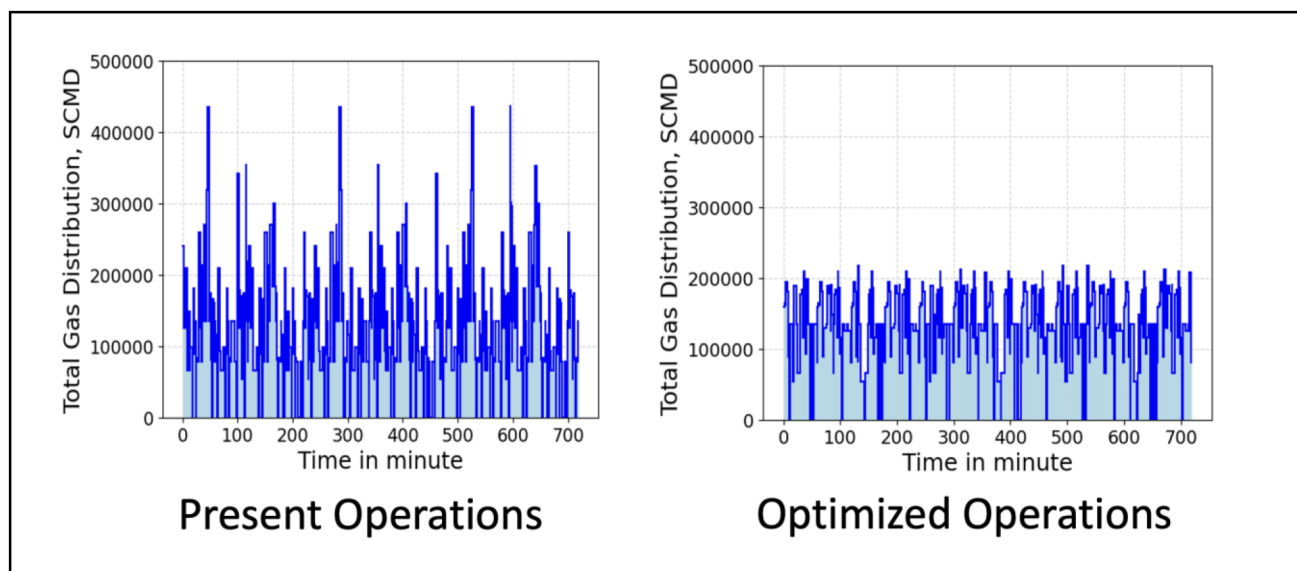


Figure 4—Peak instantaneous gas rate fluctuation with time, for Z-field of ONGC, resulting in ~50% reduction in peak instant gas load

Results

To evaluate the performance of the proposed cloud-based optimization solution, three time cycle staggering studies were carried out across ONGC's producing fields (referred to as **Field X**, **Field Y**, and **Field Z** for confidentiality). Each field had a network of intermittent gas lift (IGL) wells exhibiting operational inefficiencies due to overlapping gas injection cycles and subsequent pressure surges.

The optimization system processed well cycle inputs and delivered staggered schedules that aimed to minimize peak instantaneous gas demand. Key results from each field are summarized below.

Field X (71 IGL Wells)

- Baseline Maximum Instantaneous Gas Demand: 7.01 LSCMD
- Optimized Maximum Instantaneous Gas Demand: 3.75 LSCMD
- Peak Instantaneous Gas Demand Reduction Achieved: 46.5%
- Simultaneous Injections (Before → After): 13 → 10
- Compute Time: ~90 minutes (AWS Fargate)

Field Y (80 IGL Wells)

- Baseline Maximum Instantaneous Gas Demand: 8.23 LSCMD
- Optimized Maximum Instantaneous Gas Demand: 4.81 LSCMD
- Peak Instantaneous Gas Demand Reduction Achieved: 41.5%
- Simultaneous Injections (Before → After): 13 → 9
- Compute Time: ~95 minutes (AWS Fargate)

Field Z (29 IGL Wells)

- Baseline Maximum Instantaneous Gas Demand: 4.36 LSCMD
- Optimized Maximum Instantaneous Gas Demand: 2.17 LSCMD
- Peak Instantaneous Gas Demand Reduction Achieved: 50.2%
- Simultaneous Injections (Before → After): 4 → 2
- Compute Time: <20 minutes (AWS Fargate)

In all cases, the optimized injection cycle schedules resulted in a significant flattening of the gas demand curve, achieving consistent and measurable reductions in both peak load and injection overlaps.

Discussion

The results obtained across the three fields highlight several important insights into the effectiveness and scalability of the proposed optimization system.

Reduction in Peak Gas Demand

The system's primary objective, to minimize instantaneous gas load, was achieved with high consistency across different field sizes. Reducing the peak demand by up to 50% greatly alleviates stress on the compressor system and lift gas header, improving operational efficiency. This reduction is particularly critical in networks where daily gas supply is not the issue, but distribution timing is.

Such demand flattening is essential to maintain steady-state operating conditions in intermittent gas lift systems, where short-duration high-flow events often lead to transient failures, such as lift valve non-activation and production loss.

Injection Overlap and Pressure Stability

A significant drop in the number of concurrent gas injection events further confirms the effectiveness of the optimization framework. For example, in Field Y, the number of overlapping injection cycles dropped from 13 to 9. In Field Z, overlaps were cut in half. This staggering effect reduces the probability of sudden pressure drops across the lift gas manifold, which often cause the system to dip below the critical pressure required for valve actuation.

The proposed solution will stabilize pressure variations, resulting in enhanced efficiency of IGL wells and, consequently, increased production.

While compute times varied based on well count, with examples ranging from under 20 minutes to over 90 minutes, a deeper analysis reveals how the sequential use of Simulated Annealing, Particle Swarm Optimization, and Genetic Algorithm shapes the computational behavior. The initial SA stage rapidly explores broad regions and helps prune poor local optima, reducing the burden for PSO and GA. This cascading design leads to better convergence quality, albeit with longer runtimes than single-heuristic approaches. Internal benchmarks showed that the full three-stage sequence improved peak demand reduction by 8–35% compared to any individual algorithm, with a proportional increase of 30–40% in runtime. This trade-off favors practical field use where gas stability and production efficiency take precedence over minimal runtime.

Conclusion

This paper presents a cloud-powered optimization framework designed to enhance the operational efficiency of intermittent gas lift systems through intelligent staggering of gas injection cycles. By leveraging a sequential application of three meta-heuristic algorithms, Simulated Annealing, Particle Swarm Optimization, and Genetic Algorithm, the system effectively minimizes peak instantaneous gas demand and reduces overlapping injection cycles.

Field-scale assessments based on data from three ONGC production assets indicated the effectiveness of the proposed approach:

- Up to 50.2% reduction in peak gas load,
- Improved cycle reliability through lower overlap counts,
- Stabilized lift gas pressures, and
- Enhanced automation and scalability through cloud-native deployment.

The framework proved robust across various field configurations and computational loads. Its scalable design, integrated with AWS services such as Lambda, Fargate, SQS, S3, and SES, ensures that optimization can be achieved in minutes to hours without infrastructure bottlenecks.

As fields continue to mature and the use of IGL increases, the ability to automate and optimize time cycle scheduling will become even more critical. This solution not only addresses the current challenges but also lays the groundwork for broader AI-driven production optimization in artificial lift operations.

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